SPARK: A System for Scientifically Creative Idea Generation

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Abstract

Recently, large language models (LLMs) have shown promising abilities to generate novel research ideas in science, a direction which coincides with many foundational principles in computational creativity (CC). In light of these developments, we present an idea generation system named SPARK that couples retrieval-augmented idea generation using LLMs with a reviewer model named JUDGE trained on 600K scientific reviews from OpenReview. Our work is both a system demonstration and intended to inspire other CC researchers to explore grounding the generation and evaluation of scientific ideas within foundational CC principles. To this end, we release the annotated dataset used to train JUDGE, inviting other researchers to explore the use of LLMs for idea generation and creative evaluations.

Introduction

Automating parts of the scientific process has long been a research goal in the field of computer science and artificial intelligence (AI) (Langley 1977; Thagard 1988; Jumper et al. 2021), with computational creativity (CC) principles playing a crucial role in many parts of this endeavor. Crucially, the initial task of idea or hypothesis generation is an area that requires creative intuition but has historically been restricted to the ingenuity of human scientists.

Recent research, however, has demonstrated that large language models (LLMs) are capable of generating novel research ideas (Lee and Park 2024; Si, Yang, and Hashimoto 2024; Yang et al. 2023). In fact, a handful of fully autonomous AI systems capable of writing full-length papers at near-human quality, or workshop-length papers that successfully pass through human peer review, have been developed (Lu et al. 2024; Intology 2025). In each of these systems, idea generation remains the crucial first step.

In this paper, we present a scientific idea generation system named SPARK that generates novel scientific ideas and an evaluator model named JUDGE, trained on a 600K size dataset of expert human reviews from OpenReview, that can evaluate the creativity of SPARK's proposed ideas. Our goal is to ground recent developments in scientific idea generation with the foundational computational creativity principles that underpin much of their work, focusing on the first stage of idea generation. To facilitate reproducibility and community progress, we are releasing the full OpenReview dataset used to train our JUDGE model. Finally, to provide background context, we first start by summarizing recent developments in the use of LLMs for scientific discovery.

LLMs for Scientific Discovery

Large language models (LLMs) have been increasingly explored as tools to support creative ideation in scientific research. Early efforts such as Meta's GALACTICA utilized specialized scientific corpora to facilitate technical tasks like knowledge retrieval and hypothesis generation (Taylor, Jones, and Lee 2022). Despite strong benchmark performance, Galactica encountered significant issues with factual grounding and hallucinated references, highlighting the need for improved reliability.

Addressing grounding explicitly, SCIPIP integrated semantic retrieval and citation-awareness into idea generation, producing hypotheses informed by current literature but primarily evaluated within NLP domains (Chen, Zheng, and Gao 2023). Extending this further, SCIMON employed iterative refinement to systematically maximize novelty, though practical feasibility and experimental utility remained secondary concerns (Lee and Park 2024).

Drawing inspiration from collaborative research processes, VIRSCI orchestrated multiple LLM agents within a virtual ecosystem to simulate team-based ideation, achieving measurable improvements in idea diversity and research-trend alignment (Wang and Clark 2024). Nevertheless, like SCIMON, VIRSCI primarily focused on novelty without directly optimizing for experimental usefulness.

Other efforts aimed at automating entire research cycles include Sakana AI's AI SCIENTIST, which autonomously generates and experimentally validates hypotheses but still requires human oversight for accuracy and consistency (Lu, Zhang, and Chen 2024). Similarly, IntologyAI's ZOCHI employed multi-agent collaboration to produce workshop-accepted research papers across domains, yet it involved significant human intervention and computational resources (Intology 2025).

In this paper, we present SPARK, a demonstration aimed at grounding scientific idea generation in foundational computational creativity principles. We illustrate this approach by integrating literature retrieval, idea generation, and simulated peer-review evaluation into a unified workflow. Ad-

ditionally, we invite further exploration from the CC community by releasing the associated OpenReview dataset and non-proprietary components of our implementation.

Methods

The Spark system integrates advanced AI components designed to support computational creativity in scientific idea generation, leveraging specialized agentic systems for literature retrieval, idea synthesis, refinement, and evaluation. Figure 1 depicts the integrated process of the Spark and Xplor systems.

Literature Retrieval via Xplor

Xplor, inspired by PaperQA (Lála et al. 2023), is designed to address the specific literature retrieval and analysis needs of the Spark system. It employs sophisticated embedding techniques alongside maximal marginal relevance (MMR) for identifying, collecting, and analyzing research papers relevant to a given research question or topic (Carbonell and Goldstein 1998).

Text Embedding and Retrieval: The foundation of Xplor is OpenAI's text-embedding model, `text-embedding-ada-002`, which transforms research papers into rich 1536-dimensional semantic vectors. These embeddings facilitate accurate semantic comparisons among documents. The embeddings are indexed using the FAISS library to allow efficient nearest-neighbor searches based on cosine similarity between query embeddings and document embeddings.

Maximal Marginal Relevance: After initial retrieval, Xplor applies an MMR-based re-ranking process to optimize the selection of relevant yet diverse papers (Carbonell and Goldstein 1998). This approach iteratively selects documents with high relevance while simultaneously reducing redundancy, ensuring an informative and varied literature corpus.

Recursive Query Generation: To further refine search results, Xplor uses a recursive query generation module. In autonomous mode, Xplor initially analyzes retrieved content using an LLM employing chain-of-thought prompting to identify key themes and subtopics. This process generates increasingly targeted follow-up queries, continuing until predefined relevance metrics signal sufficient depth of literature. In interactive mode, this refinement is driven by direct user or AI feedback, enabling precise, iterative query optimization.

Idea Generation with SPARK

The Spark Idea Generator constitutes the initial creative stage of the Spark system, responsible for formulating seed research proposals grounded in rigorous prior-work analysis. It synthesizes concepts and identifies key research challenges to generate structured and contextually relevant research ideas.

The process begins with the analysis of a user-provided corpus of scientific papers. From this corpus, key terminological clusters and high-level concepts are extracted. Simultaneously, Xplor identifies emergent research gaps and themes by recursively retrieving and analyzing related papers. These outputs serve as the foundation for constructing a problem-driven research prompt.

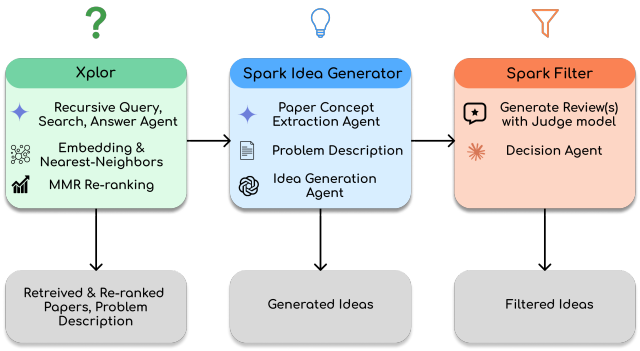

Figure 1: Spark's end-to-end pipeline: Xplor performs embedding-based, recursive literature retrieval with MMR re-ranking; the Spark Idea Generator extracts key concepts and uses chain-of-thought LLM prompting to synthesize seed research proposals; and the Spark Filter applies a judge model and decision agent for automated peer-style critique and ranking.

Given this input, a large language model agent composes a preliminary research proposal, including a title and abstract. The agent is guided to incorporate relevant domain knowledge and ensure logical coherence, while minimizing speculative or unsupported claims. The output forms the basis for subsequent refinement and evaluation, allowing the Spark system to initiate a high-quality, grounded research idea suitable for further development.

Idea Evaluation with SPARK FILTER

Spark Filter acts as the evaluative module within the Spark framework, simulating a scalable peer review process to assess the viability and quality of proposed research ideas.

In the first stage, each idea—represented by its title and abstract—is passed through a supervised, fine-tuned JUDGE model trained on a large corpus of peer reviews. This model produces multiple critiques per idea, highlighting strengths, limitations, and areas for improvement. These generated reviews aim to emulate the diverse perspectives of expert reviewers, capturing both methodological and conceptual considerations.

In the second stage, a separate decision-making model consolidates these critiques to produce a final classification of the idea as either `ACCEPT` or `REJECT`. The model applies strict evaluation criteria, synthesizing input from the prior stage to render a binary decision.

This pipeline has produced and filtered over 10,000 NLP-focused research ideas, demonstrating scalability and effectiveness for large-scale computational creativity tasks.

Evaluator Training: The JUDGE Model

Assessing the creativity and novelty of AI-generated scientific ideas demands a rigorous evaluator. Off-the-shelf

A) Inputs:

- **Problem:** "LLMs struggle with consistency"
- **Concepts:** [CoT, KGs, . . .]
- **Domain:** NLP-LLMs

B) Idea Generation Prompt:

```
Generate a novel research idea with step-by-step
reasoning based on the provided input.
```

C) LLM Response:

```
{"used":["CoT","KGs"],
 "other":["entity linking"],
 "plan":"Analyze. Wait. Try a new perspective.
 Propose KG lookups",
 "title":"KG-CoT for Fact Consistency",
 "abstract":"Integrate KGs for dynamic grounding"}
```

D) Generated Idea: Seed idea = structured JSON from (C).

Figure 2: Spark Idea Gen pipeline: (A) Inputs, (B) Idea Generation Prompt, (C) LLM Response, (D) Generated idea.

LLMs, trained via supervised fine-tuning (SFT) or reinforcement learning from human feedback (RLHF), tend to prioritize agreeableness over critical scrutiny (Ouyang et al. 2022), making them ill-suited to judge nascent research proposals. To overcome this, we introduce JUDGE, a purpose-built model trained to critique scientific ideas against established academic standards.

Data Acquisition and Preprocessing We sourced training data from OpenReview's publicly available submissions, peer reviews, and meta-reviews. An automated pipeline harvested abstract–review pairs to form the core dataset. Although we initially explored mapping full papers to their reviews, this blended conceptual appraisal with evaluations of implementation and results. Focusing on abstracts alone isolates and sharpens the model's assessment of underlying ideas.

A further challenge arises from the content of abstracts themselves. Authors often highlight strong empirical results, which can positively bias reviewer scores irrespective of the underlying idea's intrinsic novelty or merit. To mitigate this inherent bias, we introduced a data transformation step. We define an "idea abstract" ($\mathcal{A}_{idea}$) as a version of the original abstract ($\mathcal{A}_{orig}$) that deliberately omits specific results, implementation details, and performance metrics, while preserving the core problem statement, proposed approach, and claimed novelty. Similarly, an "idea review" ($\mathcal{R}_{idea}$) represents a critique focused exclusively on these conceptual aspects, abstracted away from the empirical outcomes presented in the original review ($\mathcal{R}_{orig}$).

To generate this refined dataset at scale, we employed DeepSeek-V3 as an automated annotator. Given an $(\mathcal{A}_{orig}, \mathcal{R}_{orig})$ pair, DeepSeek-V3 was prompted to perform two separate transformations: generating $\mathcal{A}_{idea}$ from $\mathcal{A}_{orig}$, and generating $\mathcal{R}_{idea}$ from $\mathcal{R}_{orig}$. The choice of DeepSeek-V3, an open-weight model, over potentially stronger proprietary models was made explicitly to enhance the reproducibility of our data generation process.

A) Input:

- **Title:** `"Enhancing Fact..."`
- **Abstract:** `"LLMs inconsistent..."`
- **Concepts:** `["CoT", "KGs", ...]`

B) Review:

```
Critique: Promising KG-CoT idea, but
abstract lacks integration detail.
Scalability with large KGs is
questionable... Needs feasibility.
```

C) Decision Prompt:

```
Review paper & reviews, output ONLY
"ACCEPT" or "REJECT".

IDEA TITLE: Enhancing Fact...
IDEA ABSTRACT: LLMs inconsistent...

REVIEWS:
REVIEW 1: Critique: Promising KG-CoT...
REVIEW 2: Strengths: Addresses key issue...
REVIEW 3: Concerns: Novelty vs prior...

Decision (ACCEPT or REJECT):
```

D) Decision:

```
REJECT
```

E) Outcome: Idea Rejected.

Figure 3: Spark Filter: Input (A), Review (B), Decision Prompt (C), Decision (D), Outcome (E).

Multi-Task Training Framework The JUDGE model was trained using a multi-task objective designed to capture both evaluative and generative aspects of scientific discourse. We formulated four distinct tasks, utilizing specialized input formats and system prompts to differentiate them:

1. **Original Review Prediction:** Mapping an original abstract to its corresponding original review, $f_{eval} : \mathcal{A}_{orig} \rightarrow \mathcal{R}_{orig}$.
2. **Idea Review Prediction:** Mapping an idea abstract to its corresponding idea review, $f_{idea_eval} : \mathcal{A}_{idea} \rightarrow \mathcal{R}_{idea}$.
3. **Original Abstract Generation:** Mapping an original review back to its corresponding original abstract, $f_{gen} : \mathcal{R}_{orig} \rightarrow \mathcal{A}_{orig}$.
4. **Idea Abstract Generation:** Mapping an idea review back to its corresponding idea abstract, $f_{idea_gen} : \mathcal{R}_{idea} \rightarrow \mathcal{A}_{idea}$.

Training encompassed these forward (evaluation) and backward (generation) mappings concurrently. This multi-task approach encourages the model to develop a deeper, bidirectional understanding of the relationship between a research idea and its critical appraisal.

Mitigating Hallucinations in AI-Driven Scientific Creativity through Citation Network Analysis and Knowledge Graph Validation

This research proposes a method for mitigating hallucinations in Large Language Models (LLMs) used for creative scientific idea generation by leveraging citation-network analysis and knowledge graphs. The approach involves constructing a knowledge graph from scientific literature—using datasets such as the *Semantic Scholar Open Research Corpus* (S2ORC)—to represent concepts and relationships within a specific scientific domain. When an LLM generates a scientific idea, its cited references are analysed within the context of this knowledge graph. Unusual combinations of cited references, identified through citation-network analysis, are flagged for further scrutiny. The content of the cited publications, processed using models like SciBERT, is then compared against the LLM's generated statements. Discrepancies between the LLM's output and the information contained in the cited references, or inconsistencies with the established knowledge graph, are flagged as potential hallucinations. However, if the unusual combination leads to a supported and insightful idea, it is considered a potentially creative contribution. This method aims to improve the reliability of AI-driven scientific discovery while exploring the possibility of leveraging controlled "hallucinations" as a source of creative insights. The research contributes to the debate surrounding AI creativity by providing a framework for evaluating the validity and potential of LLM-generated scientific ideas.

Figure 4: A research idea proposed by SPARK, using context provided by Xplor about the use of LLMs for scientific creativity.

The model was trained via continued pretraining on this structured dataset, employing Low-Rank Adaptation (LoRA) adapters to efficiently update the base model's weights (Hu et al. 2022). The training objective was to maximize the log-likelihood of the target sequence (either a review or an abstract, depending on the task) given the input sequence, formally represented as maximizing $\mathcal{L}(\theta) = \sum_{(x,y)} \log P(y|x;\theta)$, where (x, y) represents an input-output pair from one of the four tasks, and θ denotes the model parameters.

Evaluation and Model Scaling To assess the performance of JUDGE, we utilized a held-out temporal partition of the OpenReview dataset, comprising reviews and associated metadata submitted after a predefined cutoff date (e.g., post-October 2024 or from 2025 onwards). Evaluation metrics included the quality of generated reviews (assessed qualitatively and via RMSE of the final review scores in aggregate) and the accuracy of predicting explicit review scores where available.

To investigate the impact of model scale on evaluation capabilities, we trained two versions of JUDGE:

- `wintermute-tiny`: Based on the Qwen-7B model.
- `wintermute-medium`: Based on the Qwen-72B model.

This provides a roughly 10x increase in parameter count, allowing for analysis of scaling effects. Empirical validation confirmed that both `wintermute-tiny` and `wintermute-medium` consistently outperform their respective base models and other contemporary state-of-the-art LLMs on the designated evaluation tasks, demonstrating the effectiveness of our specialized training approach.

Discussion

The SPARK prototype integrates literature retrieval, multi-agent idea generation, and a simulated peer-review filter into a single workflow intended to illustrate how computational-creativity principles can be operationalized in scientific ideation. Each stage is deliberately modular: Xplor provides a retrieval layer, the Idea Generator converts retrieved context into candidate hypotheses, and the Filter simulates a scientific peer review step. By decoupling these stages, the system offers a test-bed for exploring how generation and evaluation strategies interact within the context of scientific idea generation.

Several design choices merit further exploration. First, the current filter operates only once per idea; future studies could examine closed-loop variants that feed reviewer feedback back into the generator. Second, while Xplor's retrieval is similarity-based, structured knowledge sources (e.g. ontologies) might support deeper analogical reasoning. Third, the present peer-review simulation produces textual critiques but no explicit explanatory traces; richer explanation outputs could help users understand the basis of automatic judgments. Finally, SPARK has been evaluated only in a controlled setting confined to idea generation within the field of AI, but to understand its abilities across other disciplines, broader studies are required.

To facilitate community experimentation, we release the annotated OpenReview dataset used to train the JUDGE model. We hope this proof-of-concept encourages comparative studies of retrieval methods, generation prompts, reviewer-agent designs, and creativity evaluation strategies using our system as a guide.

Conclusion

This paper has presented SPARK as a system demonstration that unifies a literature retrieval engine, an LLM-driven idea-generation module, and a peer-review–inspired filter. The implementation is offered as a concrete step towards grounding scientific idea generation in foundational computational creativity principles rather than as a definitive solution to open challenges in AI-driven discovery. By illustrating how creative generation and evaluation can be combined, we aim to inspire new directions in the computational creativity community around the generation, evaluation, and feasibility of scientific idea generation using LLM systems. We anticipate that the framework, data resources, and observations reported here will serve as a starting point for future work exploring the relationship between computational creativity and scientific idea generation.

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